ICT CORNER

Electrostatics

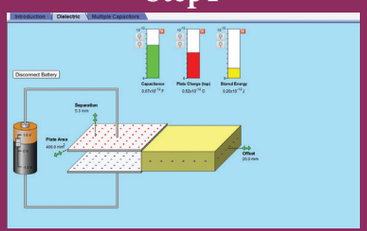
In this activity you will be able to learn about capacitor and the factors affecting capacitance.

Topic: Capacitor lab

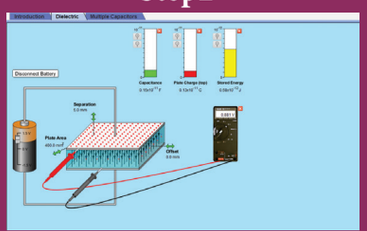
STEPS:

- Open the browser and type "phet.colorado.edu/en/simulation/legacy/capacitor-lab" in the address bar. Go to the tab 'Dielectric'.
- Change the plate area, distance between the plate and dielectric. Identify what you would maximize or minimize to make a capacitor with the greatest capacitance.
- Explore the relationships between charge, voltage, and stored energy for a capacitor. Design a capacitor system to store the greatest energy.
- Charge the capacitor with 1.0 v using the battery. Disconnect the battery. Now insert a dielectric between the plates. Discus how electric field changes in between the plates when dielectric is introduced.
- What is the effect of introducing a dielectric between plates? (Change dielectric materials)

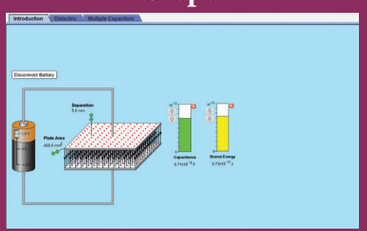
Step1



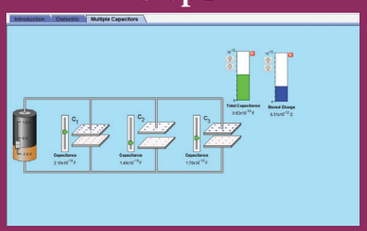
Step2



Step3



Step4



Connect capacitors parallel and series combination and find the effective capacitance.

Note:

Install Java application if it is not in your system. You can download all the phet simulation and works in off line from https://phet.colorado.edu/en/offline-access.

URL:

https://phet.colorado.edu/en/simulation/legacy/capacitor-lab

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.

12th Physics_Vol-1_Unit 1_English Medium.indd 79 08-12-2022 11:04:32